

Rational Bases and the Integer-Base Argument

A Lean-checked strategy separation for Erdős Problem #1049

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ABSTRACT

Erdős's argument for the irrationality of $\sum_{n \geq 1} c(n)b^{-n}$ at an integer base $b \geq 2$ proceeds by clearing denominators at a cut level and trapping a positive integer below 1. At a rational base r/s with $s \geq 2$ the same clearing leaves a factor s^{N+1} behind. We formalise in Lean the elementary consequence of that residue. Writing a coordinatewise corridor for the surviving arithmetic, we prove that a corridor forces $s^{N+K+1} < r(N+K)$, and that at the base $3/2$ this is impossible once the shift and the cleared window are both nonempty. We also prove the exact denominator-cleared tail recurrence $U_{N+1} = rU_N - Bc(N+1)s^{N+1}$ and the bound $2^{N+1} \leq Bc(N+1)s^{N+1}$ for $s \geq 2$ and positive data, against the exact collapse of that term to $Bc(N+1)$ at $s = 1$.

Problem #1049 is open, and nothing here decides it. The results are statements about one clearing scheme, not about the arithmetic nature of any value: we do not prove irrationality at $3/2$, at any other rational base, or in the unrestricted problem. The published height criterion that does settle a restricted family of rational bases, including $7/2$, is used as an external theorem; of its inputs we formalise only two exact arithmetic facts, and we state plainly which three inputs are not formalised. Separately we check polynomial bounds on the denominator exponents of a homogenised Padé construction, with the analytic remainder left untouched.

1 The problem

Let $t > 1$ be a rational number and let $\tau(n)$ count the divisors of n . Erdős Problem #1049 asks whether

$$F(t) = \sum_{n \geq 1} \frac{1}{t^n - 1} = \sum_{n \geq 1} \frac{\tau(n)}{t^n}$$

is irrational [2, p. 102]. The two forms agree by expanding $(t^n - 1)^{-1} = \sum_{k \geq 1} t^{-nk}$ and collecting the terms with the same exponent, the coefficient of t^{-m} being the number of divisors of m . The question is a conjecture of Chowla; Erdős proved it for every integer $t \geq 2$ [1]. Numbering and status follow Bloom's catalogue [4]. For non-integer rational t the problem is open.

Authorship and AI use. The prose in this version was generated by large-language-model agents under Will Cook's direction. Cook set the objectives and acceptance criteria, reviewed and authorised the manuscript, and is responsible for its contents. The agents are production tools, not authors. See *Statements and declarations*.

Write $t = r/s$ in lowest terms with $r > s \geq 1$, so that $s = 1$ is exactly the integer case Erdős settled. The smallest resistant explicit base is $t = 3/2$. A published height criterion of Bundschuh and Väänänen [3] settles a family of rational bases restricted by a height condition; that family contains $7/2$ and does not contain $3/2$. Between the two lies the question this note is about: what exactly stops the integer-base argument from running at $3/2$?

Relation to prior work. The statement of Problem #1049 has been formalised before, as a conjecture with an unfilled proof, in the *formal-conjectures* collection. That is a statement of the question; the present note formalises propositions about the clearing argument, and neither answers it. Prior mathematical work on the problem is Erdős’s integer-base theorem [1] and the height criterion of [3]; no claim of priority is made for anything below, which concerns the formal status of an argument rather than a new theorem about $F(t)$.

The integer-base argument, in outline. Suppose $F(b) = p/q$. Cut the series at a level N , multiply by qb^N , and subtract the cleared prefix. What remains is a positive integer, since every term of the prefix has been cleared and the tail is positive; and it is smaller than 1, since the tail of a geometric-type series is small. A positive integer below 1 does not exist. The argument is elementary and it is sensitive in exactly one place: the clearing must remove every denominator.

At $\beta = r/s$ the term $c(n)\beta^{-n}$ is $c(n)s^n/r^n$. Clearing the power of r leaves the numerator factor s^n in place. That factor is invisible when $s = 1$ and grows geometrically when $s \geq 2$. The two sections that follow record what that residue costs, first as a no-go for the clearing scheme (Section 2) and then as an exact recurrence with a lower bound on the surviving term (Section 3).

Status of everything below. *Status.* The problem treated here is open, and this note does not close it. Every statement below marked as checked is a proposition that the pinned Lean kernel accepts from the sources this note links to, with no sorry, no added axiom, and no unchecked evaluation. That is a claim about the formal statement, not about its mathematical interest, its novelty, or the original problem. The unresolved obligations are named exactly, in their own section, and none of the finite computations, reductions, or no-go results here removes one of them.

Reading map. Sections 2 and 3 are the content; Sections 4 and 5 record what is and is not formalised of two external routes. Section 6 states the remaining obligations. Linked phrases open the corresponding Lean declaration at the pinned source revision fcc93d9ade5e.

Keywords. irrationality; Lambert series; rational base; Padé approximation; Lean 4. **MSC 2020.** 11J72 (primary); 11J82, 68V20 (secondary).

2 The coordinatewise corridor

We isolate the arithmetic that the clearing scheme leaves behind, in a form that does not mention the series.

Statement	Status	Treatment here
Irrationality of $F(3/2)$	Open	Not proved, and no partial result here bears on it.
Irrationality at some rational non-integer base	Proved elsewhere	The height criterion of [3]; cited, not formalised.
Corridor forces $s^{N+K+1} < r(N+K)$	Proved here	Theorem 2.2.
No corridor at base $3/2$	Proved here	Theorem 2.4; a statement about the clearing scheme.
Cleared-tail recurrence	Proved here	Theorem 3.1, an exact identity.
Forcing term at least 2^{N+1} for $s \geq 2$	Proved here	Theorem 3.2; at $s = 1$ the term is $Bc(N+1)$.
Two arithmetic facts behind the $7/2$ criterion	Proved here	Section 4; three further inputs are not formalised.
Padé denominator-exponent bounds	Proved here	Section 5; exponent arithmetic only.
Padé remainder positivity and decay	Not treated	Section 5.

Definition 2.1 (coordinatewise corridor). Let a, b, N, K, Q, D be natural numbers. Say that (a, b, N, K, Q, D) is a *coordinatewise corridor* when

$$a > 0, \quad Q > 0, \quad D > 0, \quad D \leq N + K, \quad a^K \mid QD, \quad Qb^{N+K+1} < a^{K+1}.$$

The reading is: a and b are the numerator and denominator of the base, N is the shift, K is the width of the cleared window, Q is the accumulated clearing factor, and D is the final coefficient being cleared. The bound $D \leq N + K$ is the only property of the coefficient used; for the divisor-counting coefficient it holds because $d(n) \leq n$. The divisibility $a^K \mid QD$ is the requirement that clearing succeeded coordinatewise, and the last inequality is the tail estimate that makes the trapped integer smaller than 1. The definition is the [corridor predicate](#).

Theorem 2.2 (corridor bound). *If (a, b, N, K, Q, D) is a coordinatewise corridor, then*

$$b^{N+K+1} < a(N+K).$$

Proof. Since $QD > 0$ and $a^K \mid QD$, we have $a^K \leq QD$, and $D \leq N+K$ gives $a^K \leq Q(N+K)$. Hence

$$Qb^{N+K+1} < a^{K+1} = a^K \cdot a \leq Q(N+K) \cdot a,$$

and cancelling the positive factor Q gives the claim. $\square$

Formalised as the [power-versus-linear consequence](#).

The inequality of Theorem 2.2 is where the integer and rational cases part. At $b = 1$ it reads $1 < a(N+K)$, which holds for every $a \geq 2$ and every nonempty window; the corridor imposes no obstruction at all. At $b \geq 2$ the left side is exponential in $N+K$ and the right side is linear, so the corridor can survive only for small $N+K$. At $b = 2$ the crossing has already happened at the smallest admissible window.

Proposition 2.3. *For every natural number $x \geq 2$ we have $3x < 2^{x+1}$.*

Proof. Induction from $x = 2$, where $6 < 8$. For the step, $2^{x+1} \geq 2^2 > 3$ when $x \geq 1$, so $3(x+1) = 3x + 3 < 2^{x+1} + 2^{x+1} = 2^{x+2}$. $\square$

Formalised as the [exponential comparison](#).

Theorem 2.4 (no corridor at base $3/2$). *For all $N \geq 1$ and $K \geq 1$ and all natural Q, D , the tuple $(3, 2, N, K, Q, D)$ is not a coordinatewise corridor.*

Proof. A corridor would give $2^{N+K+1} < 3(N+K)$ by Theorem 2.2, contradicting Proposition 2.3 applied to $x = N+K \geq 2$. $\square$

Formalised as the [corridor exclusion at three halves](#).

What this section proves, and what it does not. Theorem 2.4 is a statement about Definition 2.1 and about nothing else. It says that the arithmetic left over by a literal coordinatewise clearing at base $3/2$ is inconsistent, and therefore that the integer-base argument does not transfer in that form. It does not bound the denominator of $F(3/2)$, it does not show that $F(3/2)$ is irrational, and it does not show that $F(3/2)$ is rational. It also does not rule out a clearing scheme of a different shape: the corridor fixes one divisibility pattern and one tail inequality, and an argument organised differently is not covered by the hypothesis.

3 The cleared tail and the denominator tax

Theorem 2.4 says that one scheme fails. This section identifies the quantity responsible, as an exact recurrence.

Throughout, r, s, B, F are rationals with $r \neq 0$, and $c : \mathbb{N} \rightarrow \mathbb{Q}$ is arbitrary. Define the prefix and the cleared tail state by

$$P_N = \sum_{m=0}^{N-1} \frac{c(m+1) s^{m+1}}{r^{m+1}}, \quad U_N = B r^N (F - P_N).$$

Thus P_N is the partial sum of $\sum_{n \geq 1} c(n)(s/r)^n$ through level N , and U_N is the tail of a putative value F after that level, scaled by $B r^N$. These are the [rational-base prefix](#) and the [cleared tail state](#).

Theorem 3.1 (cleared-tail recurrence). *For every N ,*

$$U_{N+1} = r U_N - B c(N+1) s^{N+1}.$$

Proof. Expanding $P_{N+1} = P_N + c(N+1)s^{N+1}/r^{N+1}$ and $r^{N+1} = r^N \cdot r$ in the definition of U_{N+1} and clearing the denominator r^{N+1} , which is nonzero, gives the identity. $\square$

Formalised as the [cleared-tail recurrence](#).

The recurrence has a linear part $r U_N$ and a forcing term $B c(N+1) s^{N+1}$. The integer-base argument works by keeping the state in a bounded window, and the forcing term is what it must absorb at each step. Its size is exactly the point of departure.

Theorem 3.2 (the forcing term). *Let s, B be natural numbers and $c : \mathbb{N} \rightarrow \mathbb{N}$, and put $G_N = B c(N + 1) s^{N+1}$.*

1. *If $s \geq 2$, $B \geq 1$ and $c(N + 1) \geq 1$, then $2^{N+1} \leq G_N$.*
2. *If $s = 1$, then $G_N = B c(N + 1)$.*

Proof. For the first part, $2^{N+1} \leq s^{N+1} = 1 \cdot s^{N+1} \leq B c(N + 1) s^{N+1}$, using $B c(N + 1) \geq 1$. The second part is the definition with $s = 1$. $\square$

Formalised as the [exponential lower bound](#) and the [integer-base collapse](#).

Reading the two parts together. At $s = 1$ the forcing term is $B c(N + 1)$, so it grows only as fast as the coefficient; for the divisor function that is $O(N^\epsilon)$ for every $\epsilon > 0$, and a bounded-state argument has room. At $s \geq 2$ the same term is at least 2^{N+1} whenever the coefficient is nonzero. This is an exact lower bound on one quantity, and it is all that is proved. It is not a proof that no bounded-state argument exists at $s \geq 2$; the theorem that one particular scheme fails is Theorem 2.4, and the bound here records the size of the term that scheme would have to absorb.

4 The height criterion at 7/2

Bundschuh and Väänänen [3] prove an irrationality criterion for a family of rational bases cut out by a height condition. The criterion applies at $\beta = 7/2$. It is an analytic theorem and it is not formalised here. Two of its elementary inputs are.

Proposition 4.1. $2^{18} < 7^7$, and $\frac{7}{18} = \frac{1}{2} - \frac{1}{9}$ in $\mathbb{Q}$.

These are the [integer power comparison](#), namely $262\,144 < 823\,543$, and the [exact rational boundary](#).

The first is the arithmetic content of the estimate $\log 2 / \log 7 < 7/18$; the passage from the integer inequality to the logarithmic one uses monotonicity of the logarithm, which is not part of the formalised statement. The second is the exact form of a margin whose other input is the strict bound $1/\pi^2 < 1/9$, that is $3 < \pi$, which is also not formalised here. Three inputs are therefore external: the logarithmic step, the bound on π , and the analytic theorem of [3] itself. We record the two exact arithmetic facts separately from the theorem precisely so that the two cannot be confused; a reader should not take Proposition 4.1 as a formalisation of the published criterion, because it is not one.

The criterion does not cover $3/2$, and the results of Sections 2 and 3 give no evidence either way about whether $F(3/2)$ is irrational.

5 Padé denominator exponents

A second external route replaces the clearing argument by explicit rational approximation. One constructs integer linear forms in 1 and $F(\beta)$ whose analytic decay outruns the height of their common denominator. For a homogenised construction over integer parameters the proposed common denominator exponent is $E_n = (3n^2 - n)/2$. Since

only doubled exponents occur below, every statement lives over $\mathbb{Z}$ and no parity book-keeping is needed; we write $\tilde{E}_n = 2E_n = 3n^2 - n$.

Proposition 5.1 (summand bound and exact gap). *Put*

$$\tilde{P}(n, k) = 2(k(n - k) + nk) + k(k - 1),$$

$$\tilde{Q}(n, m) = 2(n^2 - n) + j^2 + 2jm + j - m^2 + 3m, \quad j = n - m - 1.$$

Then, for integers n, k, m :

1. if $0 \leq k \leq n$, then $\tilde{P}(n, k) \leq \tilde{E}_n$, and the gap factors as $\tilde{E}_n - \tilde{P}(n, k) = (n - k)(3n - k - 1)$;
2. $\tilde{E}_n - \tilde{Q}(n, m) = 2(n + m(m - 1))$ identically;
3. if $n \geq 0$ and $m \geq 1$, then $\tilde{Q}(n, m) \leq \tilde{E}_n$.

Part (1) is the [summand exponent bound](#), part (2) the [exact gap identity](#), and part (3) the [maximal exponent bound](#). The gap in (2) is an identity in $\mathbb{Z}[n, m]$, so (3) follows from $m(m - 1) \geq 0$ and $n \geq 0$; the hypothesis $m \geq 1$ names the range in which the corresponding summand does not vanish, and it is not the sharpest hypothesis under which the inequality holds.

These are inequalities between polynomials in the exponents. They establish that the proposed common denominator is large enough to clear the construction, and nothing further. Positivity of the remainder, its rate of decay, and the comparison of that rate against the denominator height are the analytic obligations, and none of them is treated here. A reconstructed identity is not an irrationality measure until those are closed.

6 Complements and further questions

Problem #1049 is open. Two obligations are separated by the results above.

Problem 6.1 (the base $3/2$). Construct integer linear forms in 1 and $F(3/2)$ whose analytic decay beats the height of their common denominator. Section 5 supplies the exponent arithmetic for a homogenised construction; the remainder estimate is untouched. By Theorem 2.4, the coordinatewise clearing route is unavailable, so an argument here must be of a different shape.

Problem 6.2 (the published region). Formalise the elementary parameter inequalities of the criterion of [3], keeping the analytic theorem external and cited. Proposition 4.1 formalises two of its inputs; the logarithmic monotonicity step and the bound $3 < \pi$ remain.

The unrestricted question, which rational bases give an irrational value, remains open, and is not reduced to either problem above. What the two sections of formal content change is the status of one route, not the status of the problem: the integer-base clearing argument fails at $3/2$ for an exact and elementary reason, and the term responsible is the factor s^{N+1} in Theorem 3.1.

Statements and declarations

Artefact and data availability. The [pinned formal-source revision](#) contains the Lean sources, the fixed toolchain, and the library manifest used in the verification. This manuscript provides navigation rather than proof authority.

Declaration of generative AI use. Large-language-model agents were used throughout development to draft and revise prose, formal proofs, and software. The author set the objectives and acceptance criteria, selected and reviewed the claims, and approved the published version. The author assumes responsibility for the accuracy, interpretation, and presentation of the work. Generative systems are production tools; they are not authors and supply no independent authority. Lean checks each proof term against the fixed library version, and the sources linked here contain no proof placeholders and no project-defined axioms; Lean does not authorise the exposition, the citation choices, or the interpretation, for which the author remains responsible.

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A Guide to the formal sources

Each linked phrase opens its Lean declaration at the pinned source revision `fcc93d9ade5e`. The declarations of this note live in two modules: the corridor, the cleared-tail recurrence, and the two arithmetic certificates in the first; the Padé exponent arithmetic in the second. The link coordinates are validated against that pinned revision, so they remain correct as later work moves lines in the working tree.

References

- [1] P. Erdős, *On arithmetical properties of Lambert series*, J. Indian Math. Soc. (N.S.) **12** (1948), 63–66.
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